Project 1: Predicting Catalog Demand

Step 1: Business and Data Understanding

Provide an explanation of the key decisions that need to be made. (500 word limit)

Key Decisions:

Answer these questions

1. What decisions needs to be made?
 The key decisions to be made is whether to send out the catalogs to 250 customers in hopes that they purchase. We are trying to make predictions on how much money could be made and only want to send out the catalogs if the predictions exceed $10,000
2. What data is needed to inform those decisions?
 There are two sets of data that we must focus on. Both data sets includes the Name, Address, years the customer has been attained, etc but the customers dataset includes info about the customers history while the mailing list contains the yes or no score variable.

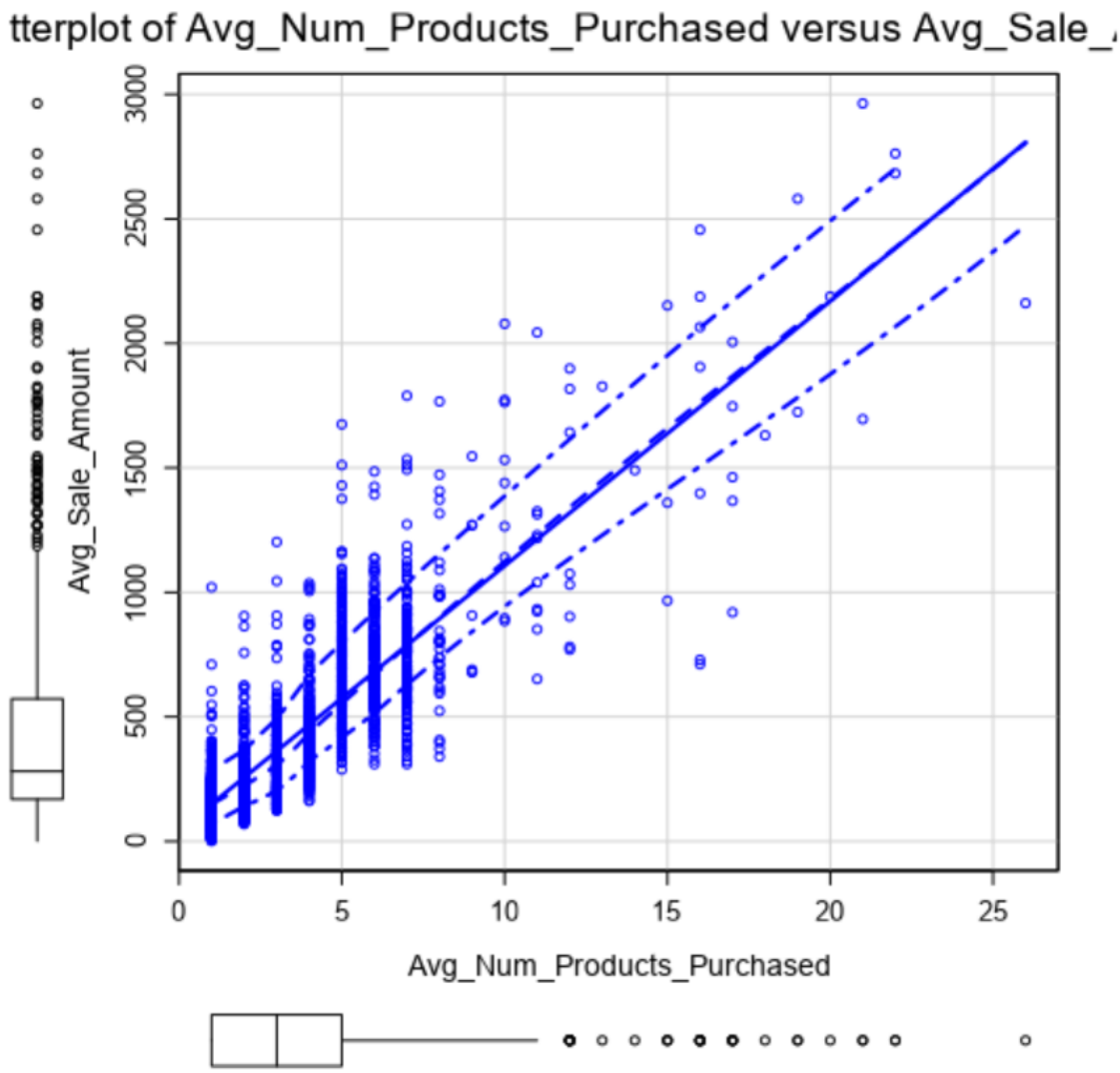
Step 2: Analysis, Modeling, and Validation

Provide a description of how you set up your linear regression model, what variables you used and why, and the results of the model. Visualizations are encouraged. (500 word limit)

Important: Use the p1-customers.xlsx to train your linear model.

At the minimum, answer these questions:

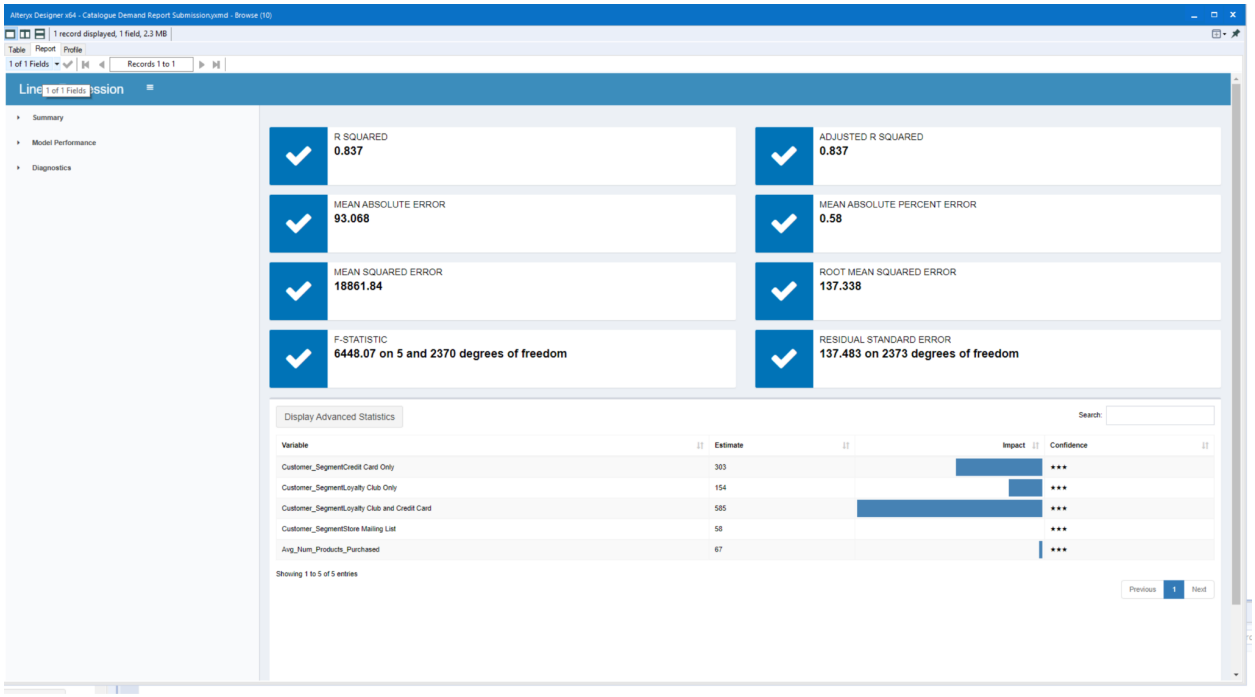
1. How and why did you select the predictor variables in your model? You must explain how your continuous predictor variables you've chosen have a linear relationship with the target variable. Please refer back to the "Multiple Linear Regression with Excel" lesson to help you explore your data and use scatterplots to search for linear relationships. You must include scatterplots in your answer.
For me the best variable to use would be the Avg_Sales variable in the customer sheet. There was no correlation between this and the other variables. I knew I needed to find a variable that I could connect with and the most logical was Average number of items purchased.



2. Explain why you believe your linear model is a good model. You must justify your reasoning using the statistical results that your regression model created. For each variable you selected, please justify how each variable is a good fit for your model by using the p-values and R-squared values that your model produced.

The p values of the variables are below 0.05 which means that the relationship between average sale amount and customer segment average number products purchased is statistically significant. The lower the better. The multiple r squared is .837 which means that the model is strong.

It's better than .7 so we can assume the model is correct.



3. What is the best linear regression equation based on the available data? Each coefficient should have no more than 2 digits after the decimal (ex: 1.28)

=303.46 - 149.36 * Customer Segment Loyalty Club Only
+281.84 * Customer Segment Loyalty Club and Credit Card
-245.42*Customer Segment Loyalty Mailing List
+0*Customer Segment Credit Only
+66.98* Avg Number Products Purchased

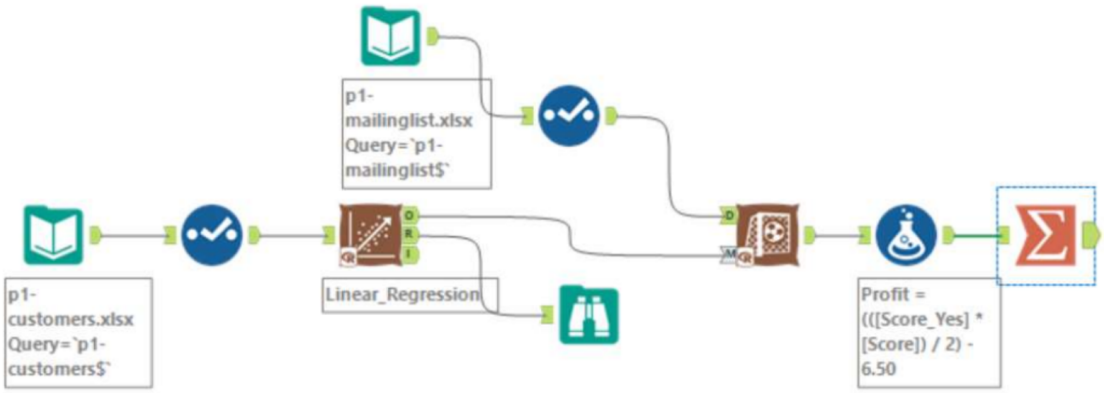
Step 3: Presentation/Visualization

Use your model results to provide a recommendation. (500 word limit)

At the minimum, answer these questions:

1. What is your recommendation? Should the company send the catalog to these 250 customers?
 The company should send out the catalogues
2. How did you come up with your recommendation? (Please explain your process so reviewers can give you feedback on your process)
I built a model on the 2300 customers and then predicted the sales of the 250 new customers. I then calculated the revenue from sending the catalogue to each new customer using the

formula (Avg_Sale Amount*Score_Yes). Then I calculated the expected profit(Sum_revenue*0.5-6.5*250) and then saw how much it was more than $10,000



3. What is the expected profit from the new catalog (assuming the catalog is sent to these 250 customers)?

$21,987.44